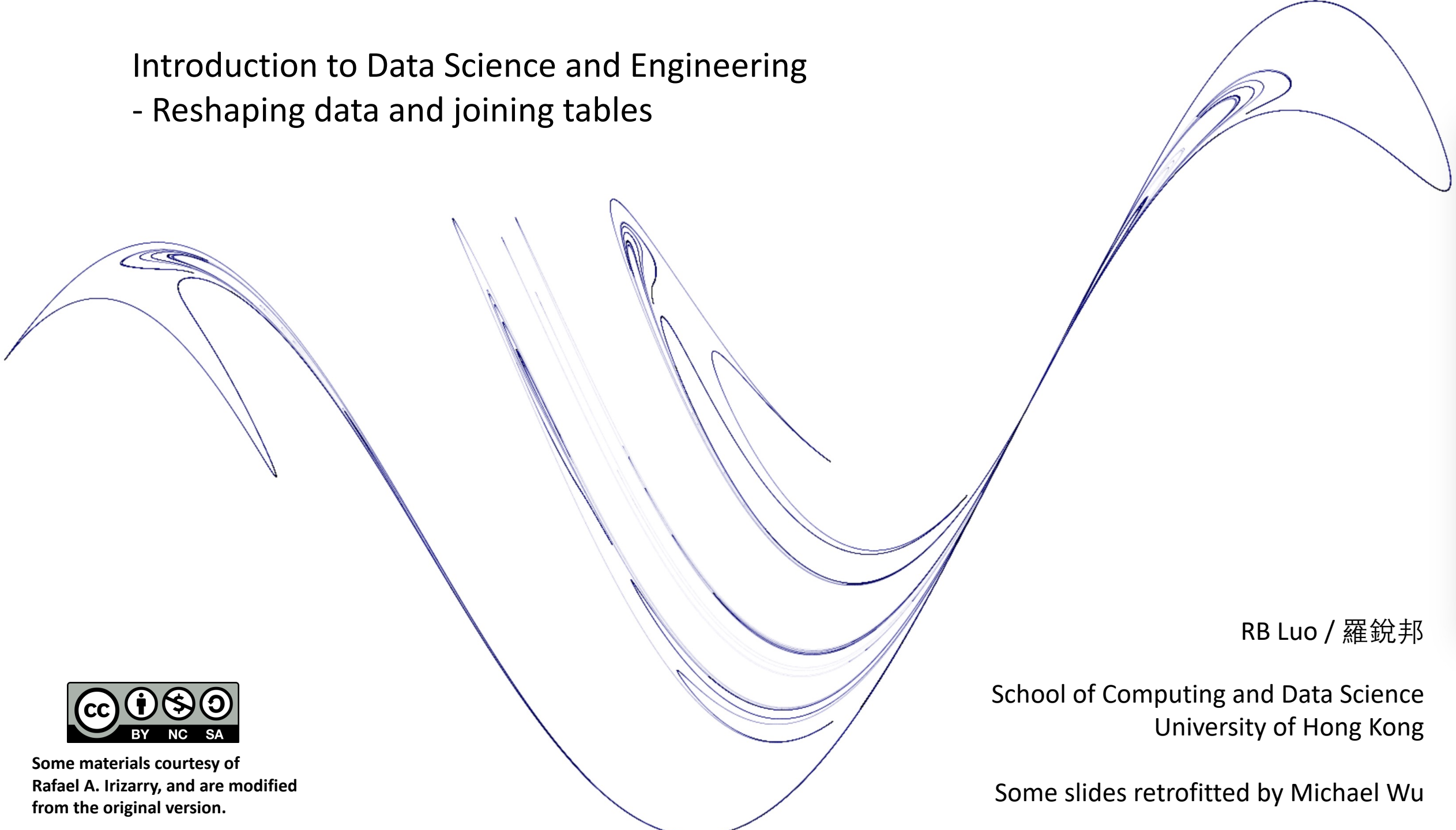
Introduction to Data Science and Engineering
- Reshaping data and joining tables



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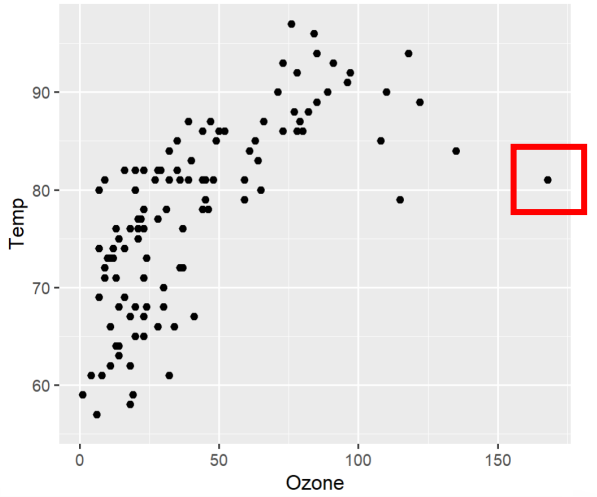
Some slides retrofitted by Michael Wu

Some materials courtesy of Rafael A. Irizarry, and are modified from the original version.

Purpose of data wrangling

- We have been using organized and tidy data: e.g., `murders`
- In real life applications, data usually comes "dirty", especially when data are extracted from web pages, tweets, and/or PDFs. In these cases, the first step is to import and **clean** the data.
  - To annotate and handle missing entries
  - To remove outliers and incorrect data
  - To standardize format
  - …




```
> data("airquality")
> mean(airquality$Ozone)
[1] NA
> sum(is.na(airquality$Ozone))
[1] 37
```

```
> airquality |> ggplot(aes(Ozone, Temp)) + geom_point()
Warning message:
Removed 37 rows containing missing values or values outside the scale range (`geom_point()`).
```

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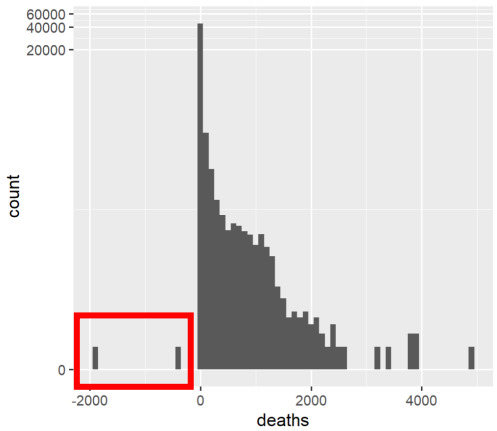
  - To standardize format
  - …

```
> covid_data <- read.csv("http://www.bio8.cs.hku.hk/comp2501/covid.csv")
> covid_data |>
+     ggplot(aes(deaths)) +
+     geom_histogram(binwidth = 100) +
+     scale_y_continuous(transform = "log1p")
```



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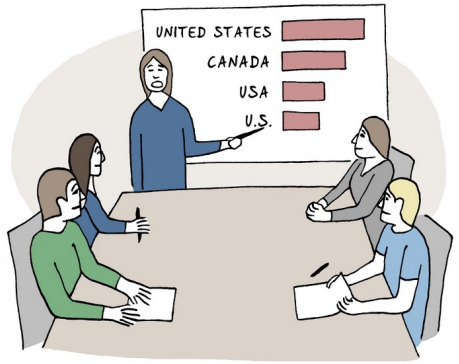
  - …



AS YOU CAN SEE, OUR TOP MARKETS ARE UNITED STATES, CANADA, USA AND THE U.S.

Dataedo /cartoon

Piotr@Dataedo

```
> data("reported_heights")
> unique(reported_heights$height)[1:50]
 [1] "75"                  "70"                  "68"                  "74"                  "61"
 [6] "65"                  "66"                  "62"                  "67"                  "72"
[11] "6"                   "69"                  "64"                  "60"                  "5' 4\""
[16] "73"                  "71"                  "66.75"               "5.3"                 "63"
[21] "70.5"                "165cm"               "511"                 "64.1732"             "68.5"
[26] "2"                   "69.6"                "5'7"                 "66.5"                "71.5"
[31] "76"                  "74.5"                "78"                  ">9000"               "5'7\""
[36] "62.5"                "5'3\""               "77"                  "68.89"               "64.173"
[41] "59"                  "67.7"                "71.7"                "70.87"               "5 feet and 8.11 inches"
[46] "5.25"                "64.57"               "51"                  "5'11"                "5.5"
```

Purpose of data wrangling

- We have been using organized and tidy data: e.g., `murders`
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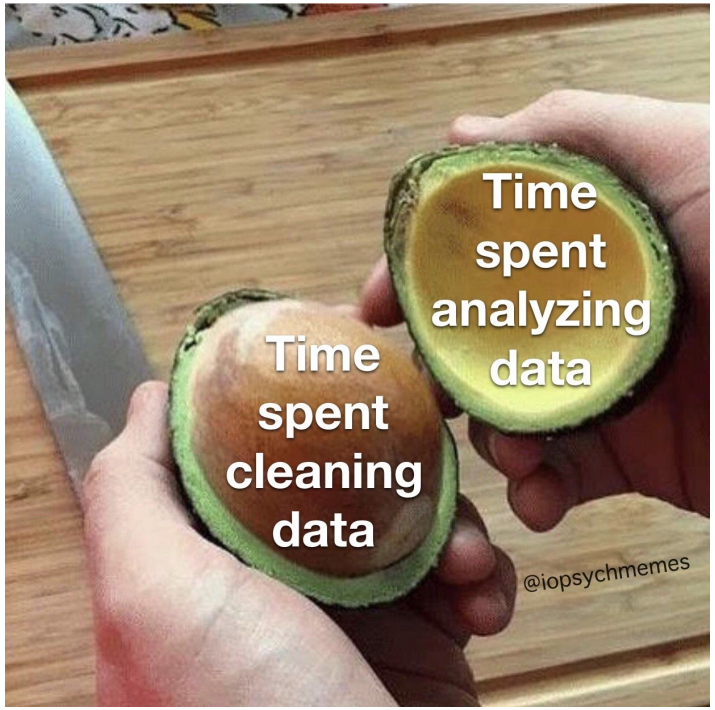


Purpose of data wrangling

- We have been using organized and tidy data: e.g., `murders`
- In real life applications, data usually comes “dirty”, especially when data are extracted from web pages, tweets, and/or PDFs. In these cases, the first step is to import and **clean** the data.
  - To annotate and handle missing entries
  - To remove outliers and incorrect data
  - To standardize format
  - …
- We will cover several common steps of the data wrangling/cleaning process:
  - Data table operations
  - String processing
  - Date/time processing
  - Web page (html) parsing

Tidy, long, and wide

Review: tidy data tables

After importing data, a common next step is to organize the data into a form that facilitates the rest of the analysis: a tidy format

*A data table is **tidy** if:*

- *each row represents **ONE** observation;*
- *each column represent a different variable available for observation*

```
> library(tidyverse)
> library(dslabs)
> path <- system.file("extdata", package="dslabs")
> filename <- file.path(path, "fertility-two-countries-example.csv")
> wide_data <- read_csv(filename)
Rows: 2 Columns: 57
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (1): country
dbl (56): 1960, 1961, 1962, 1963, 1964, 1965, 1966, 1967, 1968, 1969,...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Review: tidy data tables

After importing data, a common next step is to organize the data into a form that facilitates the rest of the analysis: a tidy format

*A data table is **tidy** if:*

- *each row represents **ONE** observation;*
- *each column represent a different variable available for observation*

```
> wide_data
# A tibble: 2 × 57
  country     `1960` `1961` `1962` `1963` `1964` `1965` `1966` `1967` `1968`
  <chr>        <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
1 Germany       2.41   2.44   2.47   2.49   2.49   2.48   2.44   2.37   2.28
2 South Korea   6.16   5.99   5.79   5.57   5.36   5.16   4.99   4.85   4.73
# … with 47 more variables: `1969` <dbl>, `1970` <dbl>, `1971` <dbl>,
#   `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>
#   `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>
#   `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>
#   `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>
#   `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, `1995` <dbl>
#   `1997` <dbl>, `1998` <dbl>, `1999` <dbl>, `2000` <dbl>
# ℹ Use `colnames()` to see all variable names
```

Wide vs Long vs Tidy

Aspect	Long Format	Wide Format	Tidy Format
Definition	Each row is an observation; fewer columns.	Variables are spread across columns.	Each variable is a column, each row an observation.
Structure	Narrow and tall (many rows).	Wide and short (many columns).	Standardized for analysis tools.
Use Case	Analysis, plotting (e.g., time-series).	Reporting, human readability.	Data analysis workflows.

Long table is not necessarily tidy:

- There could be multiple rows pointing to the same observation, but containing different variables
- No requirements for unique identifiers

Wide -> Long: `pivot_longer`

- We want to reshape the fertility dataset so that each row represents **ONE** fertility observation (one country & one year): we need three columns to store the year, country, and the observed value.
- More generally, we are splitting one row into multiple rows
  - Which columns are we splitting?
  - How to rename them?

	country	1960	1961	1962	1963	1964	1965	1966
1	Germany	2.41	2.44	2.47	2.49	2.49	2.48	2.44
2	South Korea	6.16	5.99	5.79	5.57	5.36	5.16	4.99

```
> new_tidy_data <- pivot_longer(
+     wide_data,
+     `1960`:`2015`,
+     names_to = "year",
+     values_to = "fertility",
+ )
```

	country	year	fertility
1	Germany	1960	2.41
2	Germany	1961	2.44
3	Germany	1962	2.47
4	Germany	1963	2.49
5	Germany	1964	2.49
6	Germany	1965	2.48
7	Germany	1966	2.44

What happened? backticks

If not using backticks, 1960:2015 is interpreted as a sequence

```
> new_tidy_data <- pivot_longer(wide_data, 1960:2015, names_to = "year", valu
es_to = "fertility")
Error in `build_longer_spec()`:
! Can't subset columns past the end.
i Locations 1960, 1961, 1962, …, 2014, and 2015 don't exist.
i There are only 57 columns.
Run `rlang::last_error()` to see where the error occurred.
```

- `` `1960`:`2015` `` means any columns from column named 1960 to column named 2015, including all columns in between them;
- `' '` and `""` also work, but R standard prefers using `` ` ``
- `` `` `` allows space. If a column is called "life expectancy", use `` `life expectancy` `` in `pivot_longer`

```
> new_tidy_data <- pivot_longer(wide_data, '1960':'2015', names_to = "year",
values_to = "fertility")
> new_tidy_data <- pivot_longer(wide_data, "1960":"2015", names_to = "year",
values_to = "fertility")
```

See more about using quotes in R: https://stat.ethz.ch/R-manual/R-devel/library/base/html/Quotes.html

What happened?
names_to & values_to

	country	1960	1961	1962	1963	1964	1965	1966
1	Germany	2.41	2.44	2.47	2.49	2.49	2.48	2.44
2	South Korea	6.16	5.99	5.79	5.57	5.36	5.16	4.99

names_to
values_to

	country	year	fertility
1	Germany	1960	2.41
2	Germany	1961	2.44
3	Germany	1962	2.47
4	Germany	1963	2.49
5	Germany	1964	2.49
6	Germany	1965	2.48
7	Germany	1966	2.44

A more convenient way: indexing with negative sign

Specify which column will not include in the pivot

```
> new_tidy_data <- wide_data |>
+ pivot_longer(-country, names_to = "year", values_to = "fertility")
```

- Indexing with negative number (`[-1]`) gives the same data structure, but with the specified item excluded
- Same logic here, use negative sign to indicate which column(s) to exclude from pivoting

Use correct data types

Is it all? Let's check the data frame

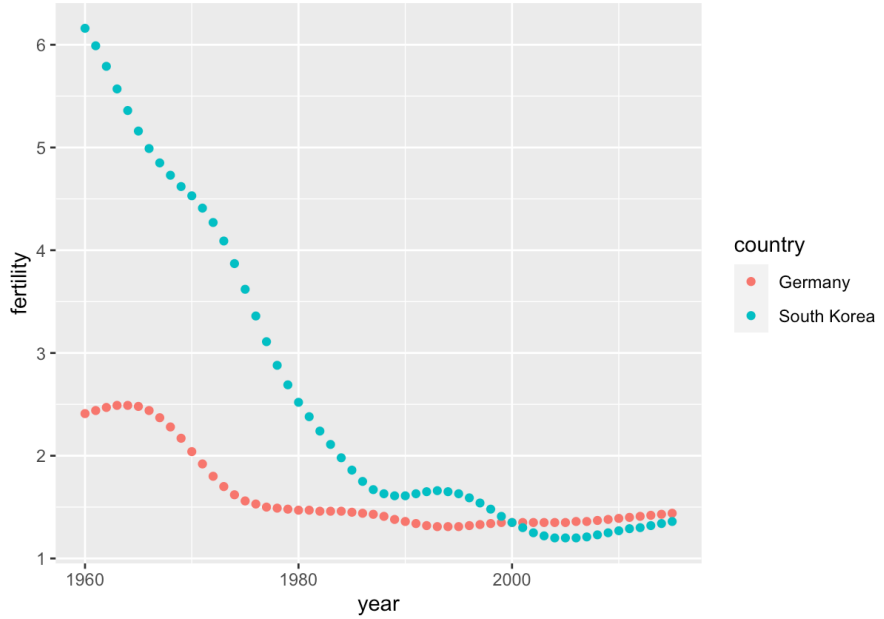
```
> str(new_tidy_data)
tibble [112 × 3] (S3: tbl_df/tbl/data.frame)
 $ country  : chr [1:112] "Germany" "Germany" "Germany" "Germany" ...
 $ year     : chr [1:112] "1960" "1961" "1962" "1963" ...
 $ fertility: num [1:112] 2.41 2.44 2.47 2.49 2.49 2.48 2.44 2.37 2.28 2.17
...
```

`pivot_longer` assumes that column names are characters To facilitate subsequent analyses, we need to convert the year column to be numbers:

```
> new_tidy_data <- wide_data |>
+ pivot_longer(-country, names_to = "year", values_to = "fertility") |>
+ mutate(year = as.integer(year))
```

Plot with tidy data

```
> new_tidy_data |>
+ ggplot(aes(year, fertility, color = country)) +
+ geom_point()
```

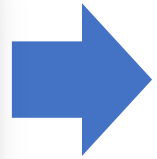


Long -> Wide: `pivot_wider`

There are also scenarios where we want to convert long data into wide data: when a long table contains multiple rows for one observation.

`pivot_wider` is the inverse of `pivot_longer`

```
> new_wide_data <- pivot_wider(
+     new_tidy_data,
+     names_from = "year",
+     values_from = "fertility",
+ )
```



```
> new_wide_data
# A tibble: 2 × 57
  country `1960` `1961` `1962` `1963` `1964` `1965` `1966` `1967` `1968`
  <chr>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
1 Germany   2.41   2.44   2.47   2.49   2.49   2.48   2.44   2.37   2.28
2 South …   6.16   5.99   5.79   5.57   5.36   5.16   4.99   4.85   4.73
# … with 47 more variables: `1969` <dbl>, `1970` <dbl>, `1971` <dbl>,
#   `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>,
#   `1984` <dbl>, `1985` <dbl>, `1986` <dbl>, `1987` <dbl>,
#   `1988` <dbl>, `1989` <dbl>, `1990` <dbl>, `1991` <dbl>,
#   `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, `1995` <dbl>, …
# ℹ Use `colnames()` to see all variable names
```

- Now we are combining multiple rows into one row, things could be tricky:
  - Two (long data) rows will be combined if (and only if) ALL other columns are the SAME.
  - All unique values in the `names_from` column of the long data table will be treated as columns for the wide data table

Long -> Wide: `pivot_wider`

```
> head(murders, 3)
    state abb region population total
1 Alabama  AL  South    4779736   135
2  Alaska  AK   West     710231    19
3 Arizona  AZ   West    6392017   232
> murders |>
+     select(region, state, population) |>
+     pivot_wider(names_from = "state", values_from = "population")
```

```
# A tibble: 4 × 52
  region       Alabama Alaska Arizona Arkansas California Colorado Connecticut Delaware `District of Columbia` Florida Georgia  Hawaii
  <fct>          <dbl>  <dbl>   <dbl>    <dbl>      <dbl>    <dbl>       <dbl>    <dbl>                 <dbl>   <dbl>   <dbl>   <dbl>
1 South        4779736     NA      NA  2915918         NA       NA          NA   897934                601723 1.97e7 9920000      NA
2 West              NA 710231 6392017       NA   37253956  5029196          NA       NA                    NA      NA      NA 1360301
3 Northeast         NA     NA      NA       NA         NA       NA     3574097       NA                    NA      NA      NA      NA
4 North Centr…      NA     NA      NA       NA         NA       NA          NA       NA                    NA      NA      NA      NA
# ℹ 39 more variables: Idaho <dbl>, Illinois <dbl>, Indiana <dbl>, Iowa <dbl>, Kansas <dbl>, Kentucky <dbl>, Louisiana <dbl>,
#   Maine <dbl>, Maryland <dbl>, Massachusetts <dbl>, Michigan <dbl>, Minnesota <dbl>, Mississippi <dbl>, Missouri <dbl>,
#   Montana <dbl>, Nebraska <dbl>, Nevada <dbl>, `New Hampshire` <dbl>, `New Jersey` <dbl>, `New Mexico` <dbl>, `New York` <dbl>,
#   `North Carolina` <dbl>, `North Dakota` <dbl>, Ohio <dbl>, Oklahoma <dbl>, Oregon <dbl>, Pennsylvania <dbl>, `Rhode Island` <dbl>,
#   `South Carolina` <dbl>, `South Dakota` <dbl>, Tennessee <dbl>, Texas <dbl>, Utah <dbl>, Vermont <dbl>, Virginia <dbl>,
#   Washington <dbl>, `West Virginia` <dbl>, Wisconsin <dbl>, Wyoming <dbl>
```

- Now we are combining multiple rows into one row, things could be tricky:
  - Two (long data) rows will be combined if (and only if) ALL other columns are the SAME.
  - All unique values in the `names_from` column of the long data table will be treated as columns for the wide data table
  - Missing entries will be filled by `NA`s

How about multiple variables stored in a wide(r) format?

```
> path <- system.file("extdata", package = "dslabs")
> filename <- "life-expectancy-and-fertility-two-countries-example.csv"
> filename <-  file.path(path, filename)
> raw_dat <- read_csv(filename)
Rows: 2 Columns: 113
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr   (1): country
dbl (112): 1960_fertility, 1960_life_expectancy, 1961_fertility, 196...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this mess
age.
```

	country	1960_fertility	1960_life_expectancy	1961_fertility	1961_life_expe
1	Germany	2.41	69.26	2.44	
2	South Korea	6.16	53.02	5.99	

A quick try with `pivot_longer`:

the table got transformed but now with one observation spanning two rows, we need to combine them (how?).

```
> raw_dat |> pivot_longer(-country)
# A tibble: 224 × 3
   country name                value
   <chr>   <chr>               <dbl>
 1 Germany 1960_fertility       2.41
 2 Germany 1960_life_expectancy 69.3
 3 Germany 1961_fertility       2.44
 4 Germany 1961_life_expectancy 69.8
 5 Germany 1962_fertility       2.47
 6 Germany 1962_life_expectancy 70.0
 7 Germany 1963_fertility       2.49
 8 Germany 1963_life_expectancy 70.1
 9 Germany 1964_fertility       2.49
10 Germany 1964_life_expectancy 70.7
# … with 214 more rows
# ℹ Use `print(n = ...)` to see more rows
```

variable

Ideally, we want to separate another "variable" column and perform `pivot_wider`

separate

The `separate` function takes three arguments: the name of the column to be separated, the names to be used for the new columns, and the separator (character that separates the variables).

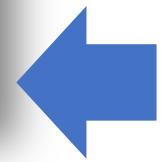
```
> raw_dat |>
+ pivot_longer(-country) |>
+ separate(name, c("year", "name"), "_")
# A tibble: 224 × 4
   country year  name      value
   <chr>   <chr> <chr>     <dbl>
 1 Germany 1960  fertility  2.41
 2 Germany 1960  life      69.3
 3 Germany 1961  fertility  2.44
 4 Germany 1961  life      69.8
 5 Germany 1962  fertility  2.47
 6 Germany 1962  life      70.0
 7 Germany 1963  fertility  2.49
 8 Germany 1963  life      70.1
 9 Germany 1964  fertility  2.49
10 Germany 1964  life      70.7
# … with 214 more rows
# ℹ Use `print(n = ...)` to see more rows
Warning message:
Expected 2 pieces. Additional pieces discarded in 112 rows [2, 4, 6, 8, 10, 1
2, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, ...].
```

You got “life” instead of “life_expectancy” (why?)

separate

The problem doesn't solve if you just keep specifying more columns

```
> raw_dat |>
+ pivot_longer(-country) |>
+ separate(name, c("year", "name1", "name2"), "_")
# A tibble: 224 × 5
   country year  name1     name2       value
   <chr>   <chr> <chr>     <chr>       <dbl>
 1 Germany 1960  fertility NA           2.41
 2 Germany 1960  life      expectancy 69.3
 3 Germany 1961  fertility NA           2.44
 4 Germany 1961  life      expectancy 69.8
 5 Germany 1962  fertility NA           2.47
 6 Germany 1962  life      expectancy 70.0
 7 Germany 1963  fertility NA           2.49
 8 Germany 1963  life      expectancy 70.1
 9 Germany 1964  fertility NA           2.49
10 Germany 1964  life      expectancy 70.7
# … with 214 more rows
# ℹ Use `print(n = ...)` to see more rows
Warning message:
Expected 3 pieces. Missing pieces filled with `NA` in 112 rows [1, 3, 5, 7,
9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, ...].
```



separate

```
> raw_dat |>
+ pivot_longer(-country) |>
+ separate(name, c("year", "name"), extra = "merge")
# A tibble: 224 × 4
   country year  name            value
   <chr>   <chr> <chr>           <dbl>
 1 Germany 1960  fertility        2.41
 2 Germany 1960  life_expectancy 69.3
 3 Germany 1961  fertility        2.44
 4 Germany 1961  life_expectancy 69.8
 5 Germany 1962  fertility        2.47
 6 Germany 1962  life_expectancy 70.0
 7 Germany 1963  fertility        2.49
 8 Germany 1963  life_expectancy 70.1
 9 Germany 1964  fertility        2.49
10 Germany 1964  life_expectancy 70.7
# … with 214 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Try `?separate`, it supports handling of multiple variables through arguments `extra` and `fill`.

A better solution is to use regular expression. We will see how to do that in the next lecture.

separate + pivot_wider

	country	1960_fertility	1960_life_expectancy	1961_fertility	1961_life_expe
1	Germany	2.41	69.26	2.44	
2	South Korea	6.16	53.02	5.99	

```
> raw_dat |>
+ pivot_longer(-country) |>
+ separate(name, c("year", "name"), extra = "merge") |>
+ pivot_wider()
# A tibble: 112 × 4
   country year  fertility life_expectancy
   <chr>   <chr>     <dbl>           <dbl>
 1 Germany 1960       2.41            69.3
 2 Germany 1961       2.44            69.8
 3 Germany 1962       2.47            70.0
 4 Germany 1963       2.49            70.1
 5 Germany 1964       2.49            70.7
 6 Germany 1965       2.48            70.6
 7 Germany 1966       2.44            70.8
 8 Germany 1967       2.37            71.0
 9 Germany 1968       2.28            70.6
10 Germany 1969       2.17            70.5
# … with 102 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Steps:

1. Separate each individual number to a row (through `pivot_longer`)
2. Specify the variable name for each row/number (through `separate`)
3. Combine numbers that belong to the same observation (through `pivot_wider`)

Joining & binding

Joining tables

With different information from multiple data sources, it is common for user to join tables according to some unique identifiers (name, id, etc.)

Suppose we want to explore the relationship between population size and electoral votes of different states. We have the population size in the `murders` dataset, and the electoral votes from the `results_us_election_2016` dataset

```
> library(dslabs)
> data(murders)
> head(murders,2)
    state abb region population total
1 Alabama  AL  South    4779736   135
2  Alaska  AK   West     710231    19
> data(polls_us_election_2016)
> head(results_us_election_2016,2)
       state electoral_votes clinton trump others
1 California              55    61.7  31.6    6.7
2     Texas               38    43.2  52.2    4.5
```

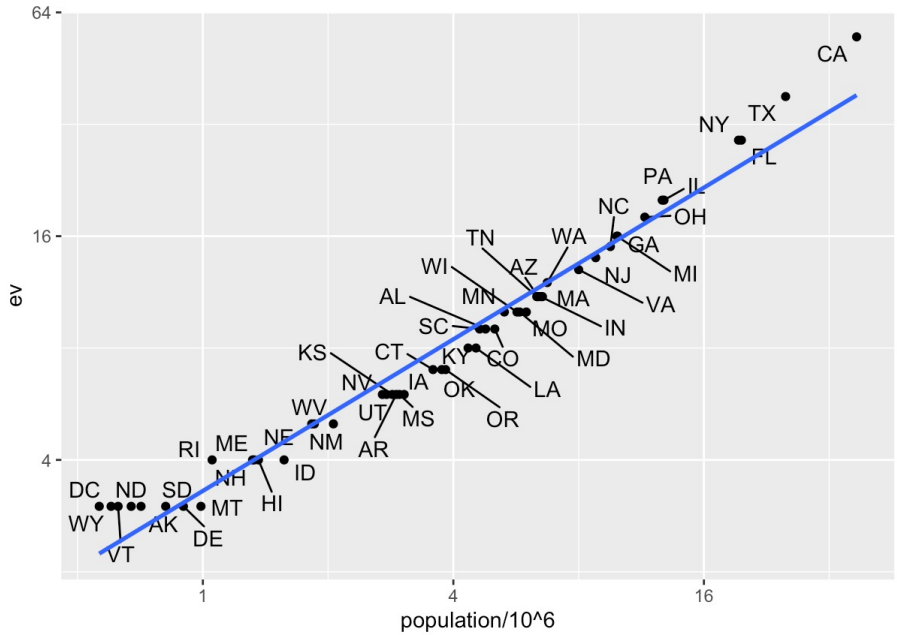
left_join

Join the two tables by state using `left_join`.

Remove the `others` column that is not needed and rename a column

```
> tab <- murders |>
+     left_join(results_us_election_2016, by = "state") |>
+     select(-others) |>
+     rename(ev = "electoral_votes")
> head(tab, n = 5)
       state abb region population total ev clinton trump
1    Alabama  AL  South    4779736   135  9    34.4  62.1
2     Alaska  AK   West     710231    19  3    36.6  51.3
3    Arizona  AZ   West    6392017   232 11    45.1  48.7
4   Arkansas  AR  South    2915918    93  6    33.7  60.6
5 California  CA   West   37253956  1257 55    61.7  31.6
```

```
> library(ggrepel)
> tab |> ggplot(aes(population/10^6, ev, label = abb)) +
+ geom_point() +
+ geom_text_repel(max.overlaps = 30) +
+ scale_x_continuous(trans = "log2") +
+ scale_y_continuous(trans = "log2") +
+ geom_smooth(method = "lm", se = FALSE)
```



We see the relationship is close to linear with about 2 electoral votes for every million persons, but with very small states getting higher ratios

Handling unmatched rows

In practice, it is not always the case that each row in one table has a matching row in another. For this reason, we have several versions of join. To demonstrate this challenge, we use a subset of the two tables. We create the tables `tab_1` and `tab_2` so that they have some states in common but not all

```
> tab_1 <- slice(murders, 1:6) |> select(state, population)
> head(tab_1)
       state population
1    Alabama    4779736
2     Alaska     710231
3    Arizona    6392017
4   Arkansas    2915918
5 California   37253956
6   Colorado    5029196
```

```
> tab_2 <- results_us_election_2016 |>
+ filter(state%in%c("Alabama", "Alaska", "Arizona",
+                   "California", "Connecticut", "Delaware")) |>
+ select(state, electoral_votes) |> rename(ev = electoral_votes)
> head(tab_2)
        state ev
1  California 55
2     Arizona 11
3     Alabama  9
4 Connecticut  7
5      Alaska  3
6    Delaware  3
```

Different types of join functions

Suppose we want a table like `tab_1`, but adding any available electoral votes from `tab_2`. For this, we use `left_join` with `tab_1` as the first argument, and `tab_2` as the second. We specify which column to use to match with the `by` argument.

```
> left_join(tab_1, tab_2, by = "state")
       state population ev
1    Alabama    4779736  9
2     Alaska     710231  3
3    Arizona    6392017 11
4   Arkansas    2915918 NA
5 California   37253956 55
6   Colorado    5029196 NA
```

States seen in `tab_1` but not `tab_2` are assigned `NA`

Different types of join functions

If we want a table like `tab_2`, but adding any available population from `tab_1`. For this, we use `right_join` with `tab_1` as the first argument and `tab_2` the second.

Notice that `right_join(tab_1, tab_2)` is equivalent to `left_join(tab_2, tab_1)`

```
> right_join(tab_1, tab_2, by = "state")
        state population ev
1     Alabama    4779736  9
2      Alaska     710231  3
3     Arizona    6392017 11
4  California   37253956 55
5 Connecticut         NA  7
6    Delaware         NA  3
```

States seen in `tab_2` but not `tab_1` are assigned `NA`

Different types of join functions

If we want to keep only the rows that have information in both tables, we use `inner_join`

```
> inner_join(tab_1, tab_2, by = "state")
       state population ev
1    Alabama    4779736  9
2     Alaska     710231  3
3    Arizona    6392017 11
4 California   37253956 55
```

Different types of join functions

If we want to keep all the rows from both tables and fill the missing parts with `NA`s, we use `full_join`

```
> full_join(tab_1, tab_2, by = "state")
        state population ev
1     Alabama    4779736  9
2      Alaska     710231  3
3     Arizona    6392017 11
4    Arkansas    2915918 NA
5  California   37253956 55
6    Colorado    5029196 NA
7 Connecticut         NA  7
8    Delaware         NA  3
```

Different types of join functions

The `semi_join` function lets us keep the part of first table that have matching rows in the second. It DOES NOT add the columns from the second table.

```
> semi_join(tab_1, tab_2, by = "state")
       state population
1    Alabama    4779736
2     Alaska     710231
3    Arizona    6392017
4 California   37253956
```

Different types of join functions

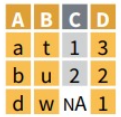
The function `anti_join` is the opposite of `semi_join`. It only keeps rows of the first table that don't have matches in the second table.

```
> anti_join(tab_1, tab_2, by = "state")
     state population
1 Arkansas    2915918
2 Colorado    5029196
```

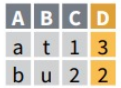
RELATIONAL DATA

Use a "**Mutating Join**" to join one table to columns from another, matching values with the rows that they correspond to. Each join retains a different combination of values from the tables.

left_join(x, y, by = NULL, copy = FALSE, suffix = c(".x", ".y"), …, keep = FALSE, na_matches = "na"**)** Join matching values from y to x.



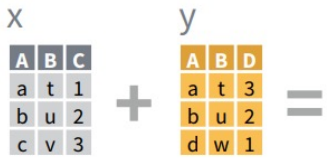
right_join(x, y, by = NULL, copy = FALSE, suffix = c(".x", ".y"), …, keep = FALSE, na_matches = "na"**)** Join matching values from x to y.



inner_join(x, y, by = NULL, copy = FALSE, suffix = c(".x", ".y"), …, keep = FALSE, na_matches = "na"**)** Join data. Retain only rows with matches.

full_join(x, y, by = NULL, copy = FALSE, suffix = c(".x", ".y"), …, keep = FALSE, na_matches = "na"**)** Join data. Retain all values, all rows.

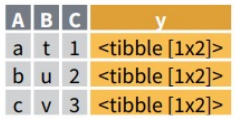
Use a "**Filtering Join**" to filter one table against the rows of another.



semi_join(x, y, by = NULL, copy = FALSE, …, na_matches = "na"**)** Return rows of x that have a match in y. Use to see what will be included in a join.

anti_join(x, y, by = NULL, copy = FALSE, …, na_matches = "na"**)** Return rows of x that do not have a match in y. Use to see what will not be included in a join.

Use a "**Nest Join**" to inner join one table to another into a nested data frame.



nest_join(x, y, by = NULL, copy = FALSE, keep = FALSE, name = NULL, …**)** Join data, nesting matches from y in a single new data frame column.

Use your cheat sheet: https://rstudio.github.io/cheatsheets/data-transformation.pdf

binding

Unlike the join functions, the binding functions do not try to match by a variable, but instead simply combine datasets. If the datasets don't match by the appropriate dimensions, yield an error

```
> bind_cols(a = 1:3, b = 4:6)
# A tibble: 3 × 2
      a     b
  <int> <int>
1     1     4
2     2     5
3     3     6
```

```
> tab_1 <- tab[1:2,]
> tab_2 <- tab[3:4,]
> bind_rows(tab_1, tab_2)
     state abb region population total ev clinton trump
1  Alabama  AL  South    4779736   135  9    34.4  62.1
2   Alaska  AK   West     710231    19  3    36.6  51.3
3  Arizona  AZ   West    6392017   232 11    45.1  48.7
4 Arkansas  AR  South    2915918    93  6    33.7  60.6
```

```
> tab_1 <- tab[, 1:3]
> tab_2 <- tab[, 4:6]
> tab_3 <- tab[, 7:8]
> new_tab <- bind_cols(tab_1, tab_2, tab_3)
> head(new_tab)
       state abb region population total ev clinton trump
1    Alabama  AL  South    4779736   135  9    34.4  62.1
2     Alaska  AK   West     710231    19  3    36.6  51.3
3    Arizona  AZ   West    6392017   232 11    45.1  48.7
4   Arkansas  AR  South    2915918    93  6    33.7  60.6
5 California  CA   West   37253956  1257 55    61.7  31.6
6   Colorado  CO   West    5029196    65  9    48.2  43.3
```

binding

Be aware of the behavior when combining rows/columns from different tables

```
> bind_cols(murders[1:5, 1:3], results_us_election_2016[1:5, 1:3])
New names:
• `state` -> `state...1`
• `state` -> `state...4`
   state...1 abb region  state...4 electoral_votes clinton
1    Alabama  AL  South California              55    61.7
2     Alaska  AK   West      Texas              38    43.2
3    Arizona  AZ   West    Florida              29    47.8
4   Arkansas  AR  South   New York              29    59.0
5 California  CA   West   Illinois              20    55.8
> bind_rows(murders[1:5, 1:3], results_us_election_2016[1:5, 1:3])
        state  abb region electoral_votes clinton
1     Alabama   AL  South              NA      NA
2      Alaska   AK   West              NA      NA
3     Arizona   AZ   West              NA      NA
4    Arkansas   AR  South              NA      NA
5  California   CA   West              NA      NA
6  California <NA>   <NA>              55    61.7
7       Texas <NA>   <NA>              38    43.2
8     Florida <NA>   <NA>              29    47.8
9    New York <NA>   <NA>              29    59.0
10   Illinois <NA>   <NA>              20    55.8
```

Set operators in R

intersect

Another set of commands useful for combining datasets are the set operators.

```
> intersect(1:10, 6:15)
[1]  6  7  8  9 10
> intersect(c("a","b","c"), c("b","c","d"))
[1] "b" "c"
```

```
> tab_1 <- tab[1:5,]
> tab_2 <- tab[3:7,]
> dplyr::intersect(tab_1, tab_2)
       state abb region population total ev clinton trump
1    Arizona  AZ   West    6392017   232 11    45.1  48.7
2   Arkansas  AR  South    2915918    93  6    33.7  60.6
3 California  CA   West   37253956  1257 55    61.7  31.6
```

union

```
> union(1:10, 6:15)
 [1]  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15
> union(c("a","b","c"), c("b","c","d"))
[1] "a" "b" "c" "d"
```

```
> tab_1 <- tab[1:5,]
> tab_2 <- tab[3:7,]
> dplyr::union(tab_1, tab_2)
        state abb    region population total ev clinton trump
1     Alabama  AL     South    4779736   135  9    34.4  62.1
2      Alaska  AK      West     710231    19  3    36.6  51.3
3     Arizona  AZ      West    6392017   232 11    45.1  48.7
4    Arkansas  AR     South    2915918    93  6    33.7  60.6
5  California  CA      West   37253956  1257 55    61.7  31.6
6    Colorado  CO      West    5029196    65  9    48.2  43.3
7 Connecticut  CT Northeast    3574097    97  7    54.6  40.9
```

setdiff

(NOT symmetric) The set difference between the first and the second argument as:

```
set(ARG1) - set(ARG2)
```

```
> setdiff(1:10, 6:15)
[1] 1 2 3 4 5
> setdiff(6:15, 1:10)
[1] 11 12 13 14 15
```

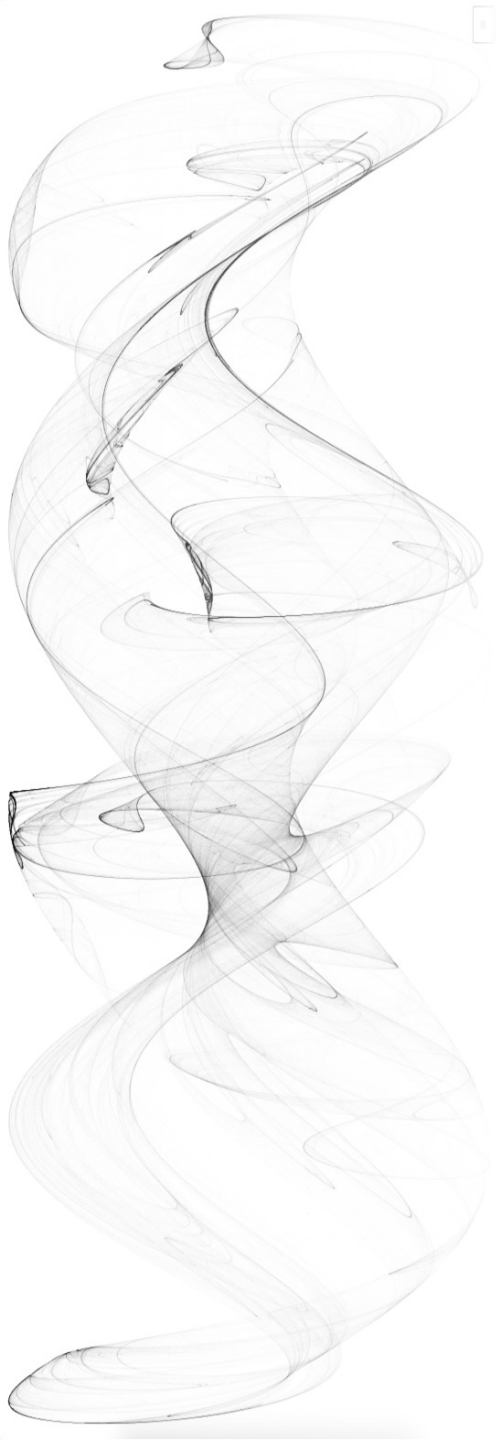
```
> tab_1 <- tab[1:5,]
> tab_2 <- tab[3:7,]
> dplyr::setdiff(tab_1, tab_2)
    state abb region population total ev clinton trump
1 Alabama  AL  South    4779736   135  9    34.4  62.1
2  Alaska  AK   West     710231    19  3    36.6  51.3
```

setequal

If the two sets are the same, regardless of order

```
> setequal(1:5, 1:6)
[1] FALSE
> setequal(1:5, 5:1)
[1] TRUE
```

```
> dplyr::setequal(tab_1, tab_2)
[1] FALSE
```



In this lecture

- Data wrangling challenges
- Tidy, long, and wide
  - `pivot_longer, pivot_wider`
  - `separate`
- Joining & binding
  - `left_join, right_join, inner_join, full_join, semi_join, anti_join`
  - `bind_rows, bind_cols`
- Set operators in R
  - `intersect, union, setdiff, setequal`